DA3406 — Data Management

Tier	A
Phase (order)	4
Do this first	DA2303, DA2402
Estimated effort	10-12 weeks
Unlocks	DA3408, DA5402

Roadmap

1. Relational model, ER, normalisation, SQL.
2. Transactions, indexing; data for AI workloads.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Database Management Systems (81 videos). IIT Madras BS *Database Management Systems* (Partha Pratim Das).

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- CMU Intro to Database Systems (15-445/645 - Fall 2025) (25 videos). CMU 15-445 *Intro to Database Systems* (Fall 2025, Andy Pavlo): how a database is built inside.

⚡ Short videos for intuition and revision

- SQL Ultimate Course (56 videos). Data with Baraa *SQL Ultimate Course*: hands-on SQL from zero to window functions.

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–10	Why DBMS, data models, relational model, intro to SQL. Install PostgreSQL.
2	13–19	Running SQL in PostgreSQL, joins, views, transactions, constraints, functions, procedures, triggers. Do 30 SQL problems (pgexercises.com).
3	20–25, 77	Relational algebra, tuple relational calculus, ER modelling, division.

Week	Lectures	What to do
4-5	26–35, 71–74, 78, 80–81	Functional dependencies, closures, keys, canonical cover, decompositions, normal forms up to 4NF. Normalise 5 schemas by hand.
6	36–40, 69	Application development: web apps, SQL from Python. Build a small Flask + Postgres app.
7	41–50, 75	Storage, file structures, indexing, B+ trees, hashing. Compare query plans with and without indexes using EXPLAIN ANALYZE.
8	51–60	Transactions, serialisability, locking, backup and recovery.
9	61–65, 76	Query processing and optimisation, NoSQL, revision.

▼ **All lectures in *Database Management Systems* (81 videos)**

1. L1: Course overview with applications prerequisites & structure of database management systems
2. L2: Why DBMS? /1 | evolution, importance, & modern applications
3. L3:Why DBMS? /2 | file based data management vs sql databases
4. L4: Introduction to DBMS/1: data abstraction, schema vs instance & models
5. L5: Introduction to DBMS/2 | data models, architecture & design principles
6. L6: Introduction to relational model/1 attributes schemas keys
7. L7: Introduction to relational model/2 with relational algebra
8. L8: Introduction to sql/1 | basics of data definition language & data manipulation language
9. L9: Introduction to SQL/2 | understanding query structure in DBMS
10. L10: Introduction to sql/3 | advanced sql concepts, set operations
11. L11: Binary numbers & character encoding part 1 | basics of binary
12. L12: Binary numbers & character encoding part 2 | ASCII unicode boolean operations
13. L13: Running SQL commands in PostgreSQL | beginner guide to psql shell
14. L14: SQL examples made easy | introduction to query building
15. L15: Intermediate sql/1 | data manipulation & nested subqueries
16. L16: Intermediate sql/2 | mastering joins & views
17. L17: Intermediate sql/3 | transactions, constraints, data types & authorization
18. L18: Advanced sql | functions, procedures & triggers explained
19. L19: Database triggers explained with case studies | functionality vs performance
20. L20: Formal relational query languages/1: introduction to relational algebra
21. L21: Formal relational query languages/2 | tuple relational calculus
22. L22: Entity-relationship model/1 | fundamentals of database design & e-r concepts
23. L23: Entity-relationship model/2 | advanced E-R diagrams & relational schema
24. L24: Entity-relationship model/3 | ternary relationships, specialization, and aggregation
25. L25: Division operation in SQL with examples
26. L26: Relational database design/1 | principles, normalization & first normal form
27. L27: Relational database design/2 | understanding functional dependencies and keys
28. L28: Relational database design/3 | functional dependency theory and database normalization

29. L29: Relational database design/4 | functional dependencies, attribute
30. L30: Relational database design/5 | dependency preservation algorithms in relational database
31. L31: Relational database design/6 | normal forms in database normalization
32. L32: Relational database design/7 | higher-level normal forms explained
33. L33: Relational database design/8 | case study in database normalization
34. L34: Relational database design/9 | multivalued dependencies (MVD) and Fourth Normal Form (4NF)
35. L35: Relational database design/10 | database design summary and handling temporal data
36. L36: Application design & development/1 | application architecture and design principles
37. L37: Application design & development/2 | web applications and database integration
38. L38: Application design & development/3 | sql and native language database connectivity
39. L39: Application design & development/4| building database applications with python and PostgreSQL
40. L40: Application design & development/5 | rapid application development and mobile app design
41. L41: Algorithms & data structures/1 | algorithms basics & complexity analysis
42. L42: Algorithms & data structures/2 | data structures | arrays,queues
43. L43: Algorithms & data structures/3 | data structures and search techniques
44. L44: Storage & file structure/1 | physical storage media in database systems
45. L45: Storage & file structure/2 | file structure | records, slotted pages
46. L46: Indexing & hashing/1 | introduction to indexing in databases
47. L47: Indexing & hashing/2 | indexing/2 | balanced trees, 2-3-4 trees
48. L48: Indexing & hashing/3 | indexing Part 3 | b plus trees, search & insert
49. L49: Indexing & hashing/4 | static hashing, dynamic hashing, and bitmap indices
50. L50: Indexing & hashing/5 | index design | locks
51. L51: Transactions/1 | acid & concurrency
52. L52: Transactions/2 | serializability | conflict serializability
53. L53: Transactions/3 | recoverability | tcl, conflict checks
54. L54: Concurrency control/1 | lock-based protocols
55. L55: Concurrency control/2 | techniques & protocols
56. L56: Backup & recovery/1 | backup/1
57. L57: Backup & recovery/2 | recovery/1
58. L58: Backup & recovery/3 | recovery techniques with logging
59. L59: Backup & recovery/4 | logical lndo & early lock release
60. L60: Backup & recovery/5 | backup/2 | redundant array of independent disks
61. L61: Query processing & optimization/1 | processing
62. L62: Query processing & optimization/2 | query optimization strategies
63. L63: Relational database systems performance & architecture
64. L64: Non-relational database management systems | nosql & cap theorem
65. L65: Widely used database management systems & course summarization
66. L66: Case study on entity relationship diagram:pharmacy database building design
67. L67: Translation of entity relation diagram into relational schema
68. L68: Entity-relationship model/2 | schema translation
69. L69: Application design & development/4 | apps with python & postgresql

70. L70: Algorithms & data structures/2 | data structures
71. L71: Lossless join decomposition | examples & conditions
72. L72: Dependency presevation
73. L73: Closure of attributes | superkeys & functional dependencies
74. L74: Super keys & candidate keys
75. L75: Indexing & hashing/3 | indexing/3 | b-tree index files
76. L76: DBMS summary tutorial | quick revision tutorial
77. L77: Tuple relational calculus | non-procedural queries
78. L78: Canonical cover | simplifying functional dependencies
79. L79: comprehensive case study on entity-relationship (ER) diagrams
80. L80: Multivalued dependency & fourth normal form
81. L81: Normalization techniques & problem solving on relational schema

Books, notes and websites

- ✅ NPTEL *Database Management System* (Partha Pratim Das)
- CMU 15-445
- Book: Silberschatz (course text); Kleppmann

Practice / proof of understanding

Design a schema + 25 SQL queries over a dataset; build a feature store table in PostgreSQL.

What your IITM course sheet says

From the PDF: DA3406

Official description: . Objectives: This course provides a comprehensive foundation in data management tailored for artificial intelligence and data analytics workloads. Building on introductory machine learning and data preprocessing, students will explore the complete lifecycle of structured and scalable data. The curriculum spans logical database design, relational algebra, and SQL, before diving into the internal mechanics of database systems, including storage, indexing, query optimization, and transaction processing. To address the demands of modern analytics, the course concludes with large-scale distributed architectures, big data paradigms (DISC and NoSQL), and emerging AI-driven database trends such as vector search and learned indexes. Upon successful completion of this course, students will be able to: • Design and Query Relational Systems: Translate conceptual data models into normalized

Official syllabus (trimmed):

2. Course contents: Module 1: Data modeling and relational foundations (3 weeks): Role of databases in AI and data analytics; Data abstraction, schemas, instances, metadata, and data independence; Conceptual modeling using entities, attributes, relationships, keys, constraints, and weak entities; ER to relational mapping; Functional dependencies and normalization, 1NF, 2NF, 3NF, BCNF; Relational model fundamentals, tuples, domains, keys, integrity constraints; Relational algebra, selection, projection, union, difference, Cartesian product, join, division, aggregation; SQL basics, DDL, DML, constraints, joins, subqueries, views, grouping, ordering, set operations. Module 2: Storage and indexing (2 weeks): DBMS storage architecture and memory hierarchy; Disk structure, pages, blocks, records, spanned and unspanned storage; File organizations, heap files, sorted files, clustered storage, hashing; Buffer manager concepts and page replacement intuition; Cost of disk I/O and basic access path analysis; Single-level and multi-level indexes; B-tree and B+ tree indexes; Static hashing and dynamic hashing, including extendible or linear hashing; Dense and sparse indexes, clustered and non-clustered indexes; Composite indexes and introductory index selection guidelines. Module 3: Query processing and transaction management (3 weeks): Query execution architecture; Evaluation o

Official textbooks: 1. Silberschatz, Abraham, Henry F. Korth, and S. Sudarshan. Database System Concepts. 7th ed. New York: McGraw-Hill Education, 2019

Official references: 1. Ramakrishnan, Raghu, and Johannes Gehrke. Database Management Systems. 3rd ed. New York: McGraw-Hill, 2002. 2. Kleppmann, Martin. Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems. Sebastopol, CA: O'Reilly Media, 2017. 3. Various research papers.